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Source: *Journal of Applied Econometrics*, Vol. 21, No. 3 (Apr., 2006), pp. 387-407

Published by: Wiley

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SMOOTHED BINARY REGRESSION QUANTILES

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SUMMARY

This paper extends results regarding smoothed median binary regression to general smoothed binary quantile regression, discusses the interpretation of the resulting estimators under alternative assumptions, and shows how they may be used to obtain semiparametric estimates of counterfactual probabilities. The estimators are applied to a model of labour force participation of married women in the USA. We find that the elasticity with respect to non-labour income is significantly negative only for women that belong to the middle of the conditional willingness-to-participate (WTP) distribution. In comparing the quantile models with parametric logit and semiparametric single-index specifications, we find that the models agree closely for women around the centre of the WTP distribution, but there are considerable disagreements as we move towards the tails of the distribution. Copyright © 2006 John Wiley & Sons, Ltd.

1. INTRODUCTION

This paper considers the extension of Horowitz's (1992) smoothed maximum score estimator of the binary choice model to quantiles other than the median. Maximum score estimation, developed and studied by Manski (1975, 1985), is equivalent to quantile regression applied to the binary choice model. Consider the model

$$\begin{aligned} Y^* &= X'\beta + U \\ Y &= 1\{Y^* \geq 0\}, \end{aligned} \quad (1)$$

where Y^* is a scalar latent continuous variable, Y is an observable binary indicator, X is a $k \times 1$ vector of explanatory variables, β is a $k \times 1$ vector of parameters, and U is a scalar random disturbance. The quantile regression latent linear specification of Koenker and Bassett (1978) is given by

$$Q_{Y^*|X}(\tau) \equiv F_{Y^*|X}^{-1}(\tau) = X'\beta_\tau, \quad \tau \in (0, 1) \quad (2)$$

where $Q(\cdot)$ and $F(\cdot)$ are, respectively, the conditional quantile and distribution functions of the latent continuous variable Y^* . This model is, for example, implied by the linear heteroskedastic location-scale model $Y^* = X'\beta + (X'\gamma)U$, $U \sim \text{i.i.d. } F_U$, yielding a quantile function for Y^* of the form (2) with $\beta_\tau = \beta + \gamma F_U^{-1}(\tau)$. Since Y^* is not observed, the model in (2) is not directly applicable here. To transform (2) into a model for the observed indicator Y we use an equivariance property of quantile functions emphasized by Powell (1984, 1986), which states that for any real, monotone increasing function $g(\cdot)$, $g(Q_{Y^*|X}(\tau)) = Q_{g(Y^*)|X}(\tau)$. This follows directly from the observation that for $g(\cdot)$ monotone increasing, $\Pr(Y^* > y|X) = \Pr(g(Y^*) > g(y)|X)$. Since the

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indicator is a monotone increasing function, $1\{Q_{Y^*|X}(\tau) > 0\} = Q_{1\{Y^* > 0\}|X}(\tau) \equiv Q_{Y|X}(\tau)$, so that the conditional quantile function of the observed indicator Y may be written as

$$Q_{Y|X}(\tau) = 1\{X'\beta_\tau \geq 0\} \quad (3)$$

This is Manski's (1975, 1985) maximum score model. Given a random sample $\{(Y_i, X_i), i = 1, \dots, n\}$ from model (1), the above reasoning suggests the *binary quantile regression estimator*

$$b_{n\tau}^* = \underset{\{b: \|b\|=1\}}{\operatorname{argmin}} \quad n^{-1} \sum_{i=1}^n \rho_\tau(Y_i - 1\{X'_i b \geq 0\}), \quad (4)$$

where $\|\cdot\|$ denotes the Euclidean norm, and $\rho_\tau(u) \equiv [\tau - 1\{u < 0\}] \cdot u$ is the asymmetric absolute loss function of Koenker and Bassett (1978). The restriction $\|b\| = 1$ normalizes the scale of the model that is not identified here. Manski (1985) defined the τ th *quantile maximum score estimator* by

$$b_{n\tau}^* = \underset{\{b: \|b\|=1\}}{\operatorname{argmax}} \quad \left\{ S_{n\tau}^*(b) = n^{-1} \sum_{i=1}^n [Y_i - (1 - \tau)] 1\{X'_i b \geq 0\} \right\} \quad (5)$$

which may be shown to be equivalent to the binary quantile regression estimator in (4).

Manski (1985) studied the asymptotic properties of the estimator and showed that it is consistent under weak conditions. Using empirical process methods, Kim and Pollard (1990) established a $O(n^{-1/3})$ rate of convergence for the estimator, and showed that the centred empirical score process converges to a Gaussian process. This limiting process, however, depends on unknown nuisance parameters that complicate its implementation for inference. To overcome these shortcomings, Horowitz (1992) proposed smoothing the score function in (5), and defined the *smoothed maximum score estimator* as the solution to the problem

$$\tilde{b}_{n\tau} = \underset{\{b: \|b\|=1\}}{\operatorname{argmax}} \quad \left\{ S_{n\tau}(b; h_n) = n^{-1} \sum_{i=1}^n [Y_i - (1 - \tau)] K\left(\frac{X'_i b}{h_n}\right) \right\} \quad (6)$$

where h_n is a (non-random) sequence of real positive constants converging to zero as n becomes large, and $K(\cdot)$ is a continuous function corresponding to the integral of an s th-order kernel ($s \geq 2$) of the type commonly used in non-parametric density and regression estimation. $S_{n\tau}(\cdot)$ is continuous and differentiable in b , and the centred smoothed estimator converges at the faster $n^{-s/(2s+1)}$ rate to an asymptotically normal random variable.

Manski (1985) defined the general binary quantile regression estimator and proved its consistency, but subsequent research, including Horowitz (1992), has focused exclusively on the median case. A number of authors, including Das (1991), Bartik *et al.* (1992), Horowitz (1993b), Gerfin (1996), Goodwin (1996), Fernandez and Rodriguez-Poo (1997), among others, have applied maximum score estimation to a variety of economic models, but have limited their attention to the median model, overlooking the potential benefits from estimating a family of quantile regression models. The purpose of this paper is to motivate and further study this extension by providing the interpretation of the estimates under alternative assumptions, extending the smoothed median results of Horowitz (1992) to families of quantile regression estimates, and indicating how such a family of estimates may be used to construct semiparametric estimates of counterfactual probabilities that no single quantile model can provide.

The general binary quantile regression model may be motivated by an array of identification, estimation and testing considerations. In many applications it happens that the number of observations in one group is much smaller than the number in the other group. For instance, in an analysis of bank failures, the number of failed banks is usually much smaller than the number of solvent banks, in a study of unemployment the number of unemployed persons is much smaller than the number of employed persons, and so on. When this happens, the index of a median model, $X_i'\beta_{.5}$, may be below zero for a large portion of individuals in the sample, and above zero for very few individuals, so the median coefficients $\beta_{.5}$ may be purely identified. It is natural in these 'rare event' cases to specify a model with respect to a quantile other than the median, so that more crossing is allowed for and identification is established.

From an estimation viewpoint, regression quantiles and their linear combinations may provide more efficient estimates than any individual quantile estimate. In rare event cases, quantiles above or below the median will most likely be more efficient, while in balanced samples, combinations of regression quantiles will improve the relative (to the median estimate) efficiency of the point estimate. In heterogenous populations, a family of quantile estimates can provide a more complete picture of how covariates affect various conditional quantiles of the latent response variable underlying the observed binary indicator.

The extension to general quantiles may also be motivated by testing considerations. Monte Carlo evidence reported by Manski and Thompson (1986) and Horowitz (1992, 1993a), among others, indicates that parametric models may produce large biases under asymmetric, fat-tailed and/or heteroskedastic distributions. Regression quantiles provide a natural context in which these violations may be tested using the heteroskedasticity test of Koenker and Bassett (1982), and the symmetry test of Newey and Powell (1987).

Binary and smoothed binary regression quantiles are solutions to difficult optimization problems that require the use of computationally expensive global optimization algorithms. The smoothed objective in (6) is most often a multimodal function, while the unsmoothed objective in (5) is also piecewise constant. In this paper we use the Goffe *et al.* (1994) simulated annealing algorithm to estimate a family of quantile models of female labour force participation in the USA. In comparing the quantile estimates with estimates from flexible parametric models and the Klein and Spady (1993) semiparametric efficient single-index model, we find that quantile models agree with the parametric and single-index models when women around the mean of the conditional willingness-to-participate (WTP) are considered, but there are considerable disagreements as we move towards the tails of the conditional WTP distribution.

2. SCALE NORMALIZATION AND INTERPRETATION OF THE ESTIMATORS

The interpretation of binary quantile regression estimates is similar to that of estimates in other quantile regression models, but is somewhat idiosyncratic due to the fact that the scale of the parameter vector β_τ in (3) is not identified. To establish identification of β_τ , $\tau \in (0, 1)$, we will follow Manski (1985) and assume that there exists a regressor whose probability distribution conditional on the other regressors is absolutely continuous with respect to Lebesgue measure, and that this regressor has support on the entire real line. Let X_k be one such regressor, define $\tilde{X} = (X_1, \dots, X_{k-1})$ to be the vector of the rest of the regressors, and let $X_1 \equiv 1$ be an intercept term. We will distinguish between two canonical cases. The first case corresponds to the situation where it is known, or may reasonably be assumed, that the error distribution $F_{U|X}(\cdot)$ is conditionally

independent of at least one of the regressors, while the second case considers the more general situation where none of the variables may reasonably be excluded from the error scale function.

Case 1. Assume that U is conditionally independent of at least one of the regressors. Without loss of generality, we may assume that X_k is one such regressor,¹ so that $F_{U|X}(\cdot) = F_{U|\tilde{X}}(\cdot)$. Although restrictive, the homoskedastic regressor assumption is justified in many applications. This is most likely to occur when X_k is determined by experimental design, as in contingent valuation studies where agents are asked if they would be willing to pay a given price ($X_k = \log\text{-price}$), or in reliability studies where tests are performed at experimentally designed times ($X_k = \log\text{-time}$).

Let $\tilde{\beta} = (\beta_1, \dots, \beta_{k-1})'$ be the coefficient vector corresponding to $\tilde{X}$, and rewrite model (1) as

$$Y = 1 \left\{ \frac{1}{\beta_k} \tilde{X}' \tilde{\beta} + X_k + U \geq 0 \right\}.$$

Since X_k does not enter the scale function, $\beta_{k,\tau}$ is constant in τ , i.e., $\beta_{k,\tau} = \beta_k$ for all $\tau \in (0, 1)$, so that

$$Q_{Y|X}(\tau) = 1 \left\{ \frac{1}{\beta_k} \tilde{X}' \tilde{\beta}_\tau + X_k \geq 0 \right\}.$$

It follows that $\tilde{\beta}_\tau$ is identified up to a common scale β_k for all $\tau \in (0, 1)$, and thus Horowitz's normalization $|\beta_{k,\tau}| = 1$ is natural in this context. Setting $F_{U|\tilde{X}}^{-1}(.5) \equiv 0$, an (up to scale) estimate of the conditional error quantile function may be obtained by

$$\frac{1}{\beta_k} F_{U|X}^{-1}(\tau) = \beta_{1,\tau} - \beta_{1,.5}.$$

Welfare measures like willingness-to-pay in contingent valuation studies are functionals of the error quantile function, so semiparametric welfare analysis is also possible if a family of quantile estimates is available.

Case 2. Now consider the more general situation where none of the regressors may reasonably be excluded from the error scale function. Although setting $|\beta_{k,\tau}| = 1$ will suffice to separately identify individual quantile models, this normalization does not facilitate comparisons of the estimation results across models. This is due to the fact that the scale of the τ th quantile model is set by $\beta_{k,\tau}$ and since this quantity is now allowed to vary across quantiles, the scale normalization may be different for different quantile regression models.

It is thus preferable in this situation to use Manski's $\|\beta_\tau\| = 1$ normalization. The intercepts, however, of the various quantile models have no direct interpretation and are not comparable across different models. In Case 1, the existence of a homoskedastic regressor made possible the interpretation of the intercepts as scaled estimates of the error quantile function, but now that all of the regressors are allowed to enter the scale function, the intercepts are not comparable across models and cannot be interpreted in the same fashion. Therefore, the normalization will have to be performed on the coefficient vector that excludes the intercept. Let β_τ^{-c} denote the τ th quantile model coefficient vector excluding the intercept, and normalize by setting $\|\beta_\tau^{-c}\| = 1$ for each $\tau \in (0, 1)$. The normalized coefficients $\beta_\tau^{-c} / \|\beta_\tau^{-c}\|$ can now be compared across different quantile models, and may be interpreted as *relative weights* that individuals in the τ th conditional

¹ If X_k is the conditionally continuous regressor but some other regressor, say X_ℓ , is the scale-exogenous one, we may estimate and obtain asymptotic standard errors under the $|\beta_k| = 1$ normalization, and then renormalize both the estimates and the standard errors so that $|\beta_\ell| = 1$.

quantile assign to their X 's. Under homoskedasticity these relative weights should remain constant across quantiles, so statistically significant movements should be interpreted as evidence of a heterogenous population in which individuals belonging to different conditional quantiles respond to their X 's differently.

3. RECOVERY OF PROBABILITIES

The idea that a family of binary quantile estimates provides a more complete picture than any single quantile estimate may alternatively be stated in terms of the completeness of the corresponding probability estimates. Let $P_{1|X} \equiv \int_{\mathbb{R}} 1\{X'\beta + U \geq 0\} dF_{U|X}$ be the conditional probability of success given a vector of characteristics X . From the definition of the τ th quantile model we have that $\Pr(Y^* \geq X'\beta_\tau | X) = 1 - \tau$, which implies that for all X

$$X'\beta_\tau \begin{matrix} \geq \\ < \end{matrix} 0 \Leftrightarrow P_{1|X} \begin{matrix} \geq \\ < \end{matrix} 1 - \tau.$$

In terms of the observed binary variable Y , the τ th quantile model restriction may be written as

$$\Pr\left(Y = 1 \mid X'\beta_\tau \begin{matrix} \geq \\ < \end{matrix} 0\right) \begin{matrix} \geq \\ < \end{matrix} 1 - \tau.$$

A single, say median, estimate separates the sample into those individuals that are above and those that are below the conditional median. In order to get a more complete picture it is thus natural to look at a *family of quantile estimates* and obtain a characterization in terms of several conditional quantiles. A change in variables, $U \rightarrow F_{U|X}^{-1}(\tau|X)$, yields

$$\begin{aligned} P_{1|X} &= \int_{\mathbb{R}} 1\{X'\beta + U \geq 0\} dF_{U|X}(U|X) \\ &= \int_0^1 1\{X'\beta_\tau \geq 0\} d\tau, \end{aligned}$$

which gives the probability of success as a function of the quantile process β_τ , $\tau \in (0, 1)$. Given the computational difficulty of estimating the entire process β_τ , we may instead usefully approximate this conditional probability over a grid of quantile estimates. Given binary quantile estimates $b_{n\tau}$ over a grid $\theta = \{\tau_1, \tau_2, \dots, \tau_m : \tau_1 < \tau_2 < \dots < \tau_m\}$, let

$$\hat{\tau}_i = \underset{\tau \in \theta}{\operatorname{argmin}} \{ \tau : X'_i b_{n\tau} \geq 0 \}, \quad i = 1, \dots, n.$$

Then an *interval* estimate of the conditional probability of success is given by

$$\hat{P}_{i,1|X_i}(\theta) = [1 - \hat{\tau}_i, 1 - \hat{\tau}_{i-1}), \quad (7)$$

where $\hat{\tau}_{i,-1}$ denotes the estimation quantile immediately preceding $\hat{\tau}_i$. For example, if estimation is performed over the grid $\theta = \{0.05, 0.10, \dots, 0.95\}$ and individual i 's quantile indices are negative for quantiles less than or equal to 0.30, and positive for quantiles 0.35 and above, then $\hat{\tau}_i = 0.35$ and $\hat{P}_{i,1|X_i}(\theta) = [0.65, 0.70)$. If all quantile indices are positive the individual belongs to the $[0.95, 1)$ bin, while if all indices are negative she belongs to the $(0, 0.05)$

probability bin. Equation (7) provides a very flexible mechanism for the semiparametric estimation of counterfactual probabilities for any set of individual characteristics X_i .

4. ASYMPTOTIC BEHAVIOUR OF THE ESTIMATORS

4.1. Consistency and Asymptotic Normality

Following Horowitz (1992), the large sample properties of the estimators are derived under the $|\beta_k| = 1$ normalization.

Assumption 1 $\tilde{\beta}_\tau \equiv (\beta_{1\tau}, \dots, \beta_{(k-1)\tau})'$ belongs to the interior of $\tilde{\mathbb{B}}$, a compact subset of $\mathbb{R}^{k-1}$.

Assumption 2 (a) The regressors $\{X_j, j = 1, \dots, k\}$ are independently distributed random n vectors with first component $X_1 \equiv 1$. (b) The support of the distribution of X is not contained in any proper linear subspace of $\mathbb{R}^k$. (c) $0 < \Pr(Y = 1|X) < 1$ for almost every X . (d) $|\beta_{k\tau}| = 1$, and for almost all $\tilde{X}$, the distribution of X_k conditional on $\tilde{X}$ has everywhere positive density with respect to Lebesgue measure.

Assumption 3 The error terms $\{U_i, i = 1, \dots, n\}$, are independent (but not necessarily identically) distributed random variables with, conditional on X , distribution functions $\{F_{U_i|X}(v), i = 1, \dots, n\}$ which are continuously differentiable with positive densities $\{f_{U_i|X}(v), i = 1, \dots, n\}$ at $v = F_{U_i|X}^{-1}(\tau)$.

Theorem 1 Under Assumptions 1–3, $b_{n\tau} \rightarrow \beta_\tau$ almost surely.

To describe the joint asymptotic distribution of the smoothed binary quantile regression estimators, let $Z_\tau \equiv \beta'_\tau X$ and $p_\tau(\cdot|\tilde{X})$ denote the density of the distribution of Z_τ conditional on $\tilde{X}$. For each integer $m > 0$ define $p_\tau^{(m)}(Z_\tau|\tilde{X}) = \partial^m p_\tau(\cdot|\tilde{X})/\partial Z_\tau^m$ whenever the derivative exists. For each β_τ such that $|\beta_{k\tau}| = 1$, let $F_\tau(\cdot|Z_\tau, \tilde{X})$ denote the distribution of U conditional on Z_τ and $\tilde{X}$.² For each positive integer m , define $F_\tau^{(m)}(-Z_\tau|Z_\tau, \tilde{X}) = \partial^m F_\tau(Z_\tau|Z_\tau, \tilde{X})/\partial Z_\tau^m$ whenever this derivative exists. For each integer $s \geq 2$, define the $(k-1) \times 1$ vector A_τ and the $(k-1) \times (k-1)$ matrices D_τ and H_τ by

$$A_\tau = -\alpha_A \sum_{m=1}^s \frac{E[F_\tau^{(m)}(0|0, \tilde{X}) p_\tau^{(s-m)}(0|\tilde{X}) \tilde{X}]}{m!(h-m)!},$$

$$D_\tau = \alpha_D \tau(1-\tau) E[\tilde{X} \tilde{X}' p_\tau(0|\tilde{X})] \quad \text{and} \quad H_\tau = E[\tilde{X} \tilde{X}' F_\tau^{(1)}(0|0, \tilde{X}) p_\tau(0|\tilde{X})],$$

where $\alpha_A = \int_{-\infty}^{\infty} v^s K^{(1)}(v) dv$ and $\alpha_D = \int_{-\infty}^{\infty} [K^{(1)}(v)]^2 dv$. We will make the following additional assumptions.

Assumption 4 $nh_n \rightarrow \infty$ and $(\log n)/(nh_n^4) \rightarrow 0$, as $n \rightarrow \infty$.

² $F_\tau(\cdot|Z_\tau, \tilde{X})$ denotes the conditional distribution function of U under the τ th quantile assumption, and is such that $F_\tau(0|0, \tilde{X}) = \tau$. Similarly, $p_\tau(\cdot|\tilde{X})$ denotes the conditional density of Z_τ under the τ th quantile model.

Assumption 5 (a) $|K(v)| < M$ for all v in the support of K and some finite constant M . (b) $\lim_{v \rightarrow \infty} K(v) = 1$, $\lim_{v \rightarrow -\infty} K(v) = 0$. (c) K is twice differentiable everywhere, $|K^{(1)}(v)|$ and $|K^{(2)}(v)|$ are uniformly bounded, and each of the following integrals is finite: $\int_{-\infty}^{\infty} [K^{(1)}(v)]^4 dv$, $\int_{-\infty}^{\infty} [K^{(2)}(v)]^2 dv$, $\int_{-\infty}^{\infty} |v^2 K^{(2)}(v)| dv$. (d) For every integer $s \geq 2$ and each integer m , $1 \leq m \leq s$, $\int_{-\infty}^{\infty} |v^m K^{(1)}(v)| dv$ is finite, and $K^{(1)}(v)$ is an s th-order kernel. (e) For every integer m , $0 \leq m \leq s$, any $\eta > 0$, and any sequence $h_n \rightarrow 0$,

$$\lim_{n \rightarrow \infty} h_n^{m-s} \int_{|h_n v| > \eta} |v^m K^{(1)}(v)| dv = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} h_n^{-1} \int_{|h_n v| > \eta} |K^{(2)}(v)| dv = 0.$$

Assumption 6 (a) For each integer m , $0 \leq m \leq s-1$, all Z_τ in a neighbourhood of 0, almost every $\tilde{X}$, and some finite constant M , $p_\tau^{(m)}(Z_\tau|\tilde{X})$ exists and is a continuous function of Z_τ satisfying $|p_\tau^{(m)}(Z_\tau|\tilde{X})| < M$. (b) For each integer m , $1 \leq m \leq s-1$, all Z_τ in a neighbourhood of 0, almost every $\tilde{X}$, and some finite constant M , $F_\tau^{(m)}(-Z_\tau|Z_\tau, \tilde{X})$ exists and is a continuous function of X satisfying $|F_\tau^{(m)}(Z_\tau|\tilde{X})| < M$.

Assumption 7 The absolute moments $E|\tilde{X}|$, $E|\tilde{X}'\tilde{X}|$, $E|\tilde{X}\tilde{X}'\tilde{X}\tilde{X}'|$ are finite, and the matrix H_τ is negative definite.

Let $\theta = (\tau_1, \tau_2, \dots, \tau_m)$ be a vector of m quantiles such that $0 < \tau_1 < \tau_2 < \dots < \tau_m < 1$, $\gamma_\theta = (\beta'_{\tau_1}, \beta'_{\tau_2}, \dots, \beta'_{\tau_m})'$ be a $(km \times 1)$ vector of population regression quantiles and $c_{n\theta} = (b'_{n\tau_1}, b'_{n\tau_2}, \dots, b'_{n\tau_m})'$ be the corresponding vector of sample smoothed regression quantiles. Define $\tilde{D}_{\tau_j \tau_\ell} = \alpha_D(\min\{\tau_j, \tau_\ell\} - \tau_j \tau_\ell) E[\tilde{x} \tilde{x}' p_{\tau_j}(0|\tilde{x}) p_{\tau_\ell}(0|\tilde{x})]$, and let

$$D_{\tau_j \tau_\ell} = \begin{cases} D_\tau & \text{if } \tau_j = \tau_\ell = \tau, \\ \tilde{D}_{\tau_j \tau_\ell} & \text{if } \tau_j \neq \tau_\ell. \end{cases} \quad (8)$$

Observe that $\tilde{D}_{\tau_j \tau_\ell}$ does not reduce to D_τ in the case where $\tau_j = \tau_\ell = \tau$, thus the separate definitions above. Also define the $[m(k-1) \times 1]$ vector $\Lambda_\theta = ((H_{\tau_1}^{-1} A_{\tau_1})', \dots, (H_{\tau_m}^{-1} A_{\tau_m})')'$, and the $[m(k-1) \times m(k-1)]$ matrix $\Sigma_\theta = \{H_{\tau_j}^{-1} D_{\tau_j \tau_\ell} H_{\tau_\ell}^{-1}\}_{j, \ell=1, \dots, m}$. The following theorem describes the joint asymptotic distribution of a set of smoothed binary regression quantiles.

Theorem 2 Let Assumptions 1–7 hold for some finite integer $s \geq 2$.

(a) If nh_n^{2s+1} has a finite limit λ as $n \rightarrow \infty$,

$$(nh_n)^{1/2}(\tilde{c}_{n\theta} - \tilde{\gamma}_\theta) \xrightarrow{d} \mathcal{N}_{m(k-1)}(-\lambda^{1/2} \Lambda_\theta, \Sigma_\theta).$$

(b) Let $h_n = (\lambda/n)^{1/(2s+1)}$ with $0 < \lambda < \infty$. Then $AMSE \equiv E(\tilde{c}_{n\theta} - \tilde{\gamma}_\theta)'(\tilde{c}_{n\theta} - \tilde{\gamma}_\theta)$ is minimized by setting $\lambda = \lambda_\theta^* \equiv \text{trace}(\Sigma_\theta)/(2s\Lambda_\theta' \Lambda_\theta)$, in which case

$$n^{s/(2s+1)}(\tilde{c}_{n\theta} - \tilde{\gamma}_\theta) \xrightarrow{d} \mathcal{N}_{m(k-1)}[-(\lambda_\theta^*)^{s/(2s+1)} \Lambda_\theta, (\lambda_\theta^*)^{-1/(2s+1)} \Sigma_\theta].$$

The following corollary describes the asymptotic distribution of a single smoothed binary quantile estimate.

Corollary 1 Under the conditions of Theorem 2, if $\theta = \{\tau\}$, a singleton,

$$n^{s/(2s+1)}(\tilde{b}_{n\tau} - \tilde{\beta}_\tau) \xrightarrow{d} \mathcal{N}_{k-1}[-(\lambda_\tau^*)^{s/(2s+1)}H_\tau^{-1}A_\tau, (\lambda_\tau^*)^{-1/(2s+1)}H_\tau^{-1}D_\tau H_\tau^{-1}],$$

where $\lambda_\tau^* = \text{trace}(H_\tau^{-1}D_\tau H_\tau^{-1})/(2sA_\tau' H_\tau^{-1}A_\tau)$.

Under i.i.d. errors, the slope parameters of $b_{n\tau}$ estimate the same quantity irrespective of the choice of τ , so one would like to know if some quantile models may provide a more efficient estimate than others. Under i.i.d. sampling, $F_\tau^{(1)}(0|0, \tilde{X}) = F_\tau^{(1)}(0) = f(F^{-1}(\tau))$ w.p. 1, and the covariance matrix in the asymptotic distribution in Corollary 1 simplifies to $(\lambda_\tau^*)^{-1/(2s+1)}\omega_\tau \cdot (E[p_\tau(0|\tilde{X})\tilde{X}\tilde{X}'])^{-1}$, where $\omega_\tau = \alpha_D\tau(1-\tau)/f(F^{-1}(\tau))^2$. The efficiency of the estimate therefore depends: (a) on the accuracy with which the τ th quantile can be estimated (reflected in the ω_τ part), (b) on the X 's themselves (through the XX' term) and (c) on the probability that the index $X'b_{n\tau}$ equals zero (approximated by the density of Z_τ evaluated at zero). The first two terms are usual in quantile regression, while the last, $p_\tau(0|\tilde{X})$, is characteristic of the binary model. Although the probability that the index $X'b_{n\tau}$ equals zero depends on the structure of the data, it also depends on the choice of τ . In the case of disproportionate sampling (i.e., many more zeros than ones or vice versa), one may increase this probability by choosing an appropriate quantile, i.e., a quantile above the median if the sample has many more zeros than ones, and a quantile below the median if the ones are many more than the zeros. In a sample with, say, many more zeros than ones, the information is mainly contained in the ones, so choosing a high quantile places more weight on the rare $Y = 1$ event, and centres estimation around the informative part of the sample.

4.2. Consistent Estimation of Nuisance Parameters

A consistent estimator of D_τ may be obtained by the outer product of the first derivative of the sample score function, and similarly a consistent estimator for H_τ may be obtained by the Hessian of the sample score function. Let

$$T_{n\tau}(b; h_n) = \frac{\partial S_{n\tau}(b; h_n)}{\partial \tilde{b}} = (nh_n)^{-1} \sum_{i=1}^n [Y_i - (1 - \tau)] \tilde{X}_i K^{(1)} \left(\frac{X_i' b}{h_n} \right), \quad (9)$$

and

$$H_{n\tau}(b; h_n) = \frac{\partial^2 S_{n\tau}(b; h_n)}{\partial \tilde{b} \partial \tilde{b}'} = (nh_n^2)^{-1} \sum_{i=1}^n [Y_i - (1 - \tau)] \tilde{X}_i \tilde{X}_i' K^{(2)} \left(\frac{X_i' b}{h_n} \right) \quad (10)$$

be the vector of first and the matrix of second derivatives of the objective function, respectively. For some $0 < \delta < 1$, let $h_{n\delta} = O(n^{-\delta/(2s+1)})$ and define

$$\hat{A}_{n\tau} = (h_{n\delta})^{-s} T_{n\tau}(b_{n\tau}; h_{n\delta}) = (nh_{n\delta}^{s+1})^{-1} \sum_{i=1}^n [Y_i - (1 - \tau)] \tilde{X}_i K^{(1)} \left(\frac{X_i' b_{n\tau}}{h_{n\delta}} \right) \quad (11)$$

to be an estimator of A_τ . To obtain an estimator for D_τ consider $t_{n\tau i}(b_{n\tau}; \sigma_n) = \sigma_n^{-1} [Y_i - (1 - \tau)] \tilde{X}_i K^{(1)}(X_i' b_{n\tau}/h_n)$, and note that $T_{n\tau}(b; \sigma_n) = n^{-1} \sum_i t_{n\tau i}(b_{n\tau}; \sigma_n)$, so that a consistent estimator of D_τ is given by

$$(h_n/n) \sum_{i=1}^n t_{n\tau i}(b_{n\tau}; h_n) t_{n\tau i}(b_{n\tau}; h_n)' = (nh_n)^{-1} \sum_{i=1}^n [Y_i - (1 - \tau)]^2 \tilde{X}_i \tilde{X}_i' \left[K^{(1)} \left(\frac{X_i' b_{n\tau}}{h_n} \right) \right]^2.$$

Under the τ th quantile model, $E[Y - (1 - \tau)|\tilde{X}]^2 = \tau(1 - \tau)$. Using this, the definition of α_D and the assumption that $K^{(1)}(\cdot)$ integrates to 1, we may define an estimator of D_τ as

$$\hat{D}_{n\tau} = \alpha_D \tau(1 - \tau)(nh_n)^{-1} \sum_{i=1}^n \tilde{X}_i \tilde{X}_i' K^{(1)}\left(\frac{X_i' b_{n\tau}}{h_n}\right). \quad (12)$$

Similarly, to estimate $\tilde{D}_{n\tau_j\tau_\ell}$ we may use

$$\hat{\tilde{D}}_{n\tau_j\tau_\ell} = (\min\{\tau_j, \tau_\ell\} - \tau_j\tau_\ell)(nh_n)^{-1} \sum_{i=1}^n \tilde{X}_i \tilde{X}_i' K^{(1)}\left(\frac{X_i' b_{n\tau_j}}{h_n}\right) K^{(1)}\left(\frac{X_i' b_{n\tau_\ell}}{h_n}\right), \quad (13)$$

and

$$\hat{D}_{n\tau_j\tau_\ell} = \begin{cases} \hat{D}_{n\tau} & \text{for } \tau_j = \tau_\ell = \tau, \\ \hat{\tilde{D}}_{n\tau_j\tau_\ell} & \text{for } \tau_j \neq \tau_\ell. \end{cases} \quad (14)$$

Finally, an estimator for H_τ is given by $\hat{H}_{n\tau} = H_{n\tau}(b_{n\tau}; h_n)$.

Theorem 3 Under the assumptions of Theorem 2, $\hat{A}_{n\tau} \xrightarrow{p} A_\tau$, $\hat{D}_{n\tau_j\tau_\ell} \xrightarrow{p} D_{\tau_j\tau_\ell}$ and $\hat{H}_{n\tau} \xrightarrow{p} H_\tau$. It follows that $\hat{\Lambda}_\theta \equiv ((\hat{H}_{\tau_1}^{-1}\hat{A}_{\tau_1})', \dots, (\hat{H}_{\tau_m}^{-1}\hat{A}_{\tau_m})')' \xrightarrow{p} \Lambda_\theta$ and $\hat{\Sigma}_\theta \equiv \{\hat{H}_{\tau_j}^{-1}\hat{D}_{\tau_j\tau_\ell} \hat{H}_{\tau_\ell}^{-1}\}_{j,\ell=1,\dots,m} \xrightarrow{p} \Sigma_\theta$.

4.3. Testing the Quantile Regression Specification

While in many applications the linear quantile specification may be an adequate and parsimonious approximation to the true conditional quantile function, its adequacy should be checked. The τ th quantile assumption implies that in the population $\Pr(Y^* \geq X'\beta_\tau|X) = 1 - \tau$, and this condition provides a benchmark upon which to test the adequacy of the τ th quantile model for any $\tau \in (0, 1)$. The testable implication is the 'single-crossing' condition

$$\Pr(Y = 1|X'\beta_\tau = 0) = 1 - \tau. \quad (15)$$

The conditional probability $\Pr(Y = 1|X'\beta_\tau)$ may be consistently estimated by non-parametrically regressing Y on $X'b_{n\tau}$, while a uniform confidence band for the estimate may be obtained from theorem 4.3.1 of Härdle (1990, pp. 116–117) (see also Horowitz, 1993b). The bandwidth for non-parametric estimation may be chosen by standard cross-validation methods. A specification test for the τ th quantile model at the α level of significance may then be implemented by checking that the $(1 - \alpha)$ uniform confidence band includes $(1 - \tau)$ when the non-parametric probability estimate is evaluated at $X'b_{n\tau} = 0$.

5. EMPIRICAL APPLICATION: LABOUR FORCE PARTICIPATION OF MARRIED WOMEN

This section presents an empirical application to a model of the labour force participation decision of married women in the USA. Our concern will be with modelling the (indirect) latent utility differential between working and not working, which may be interpreted as a

propensity or *willingness-to-participate* (WTP) index. Large positive values of this index signify high willingness, large negative values signify high unwillingness, and values around zero indicate relative indifference between the two alternatives.

5.1. The Models

Let Y^* denote the latent WTP. Along with the semiparametric smoothed binary regression quantiles (SBRQ), we will also consider three parametric logit models, as well as the semiparametric Klein and Spady (1993) single-index model. The logit and single-index estimators all maximize the familiar log-likelihood of the binary response model,

$$\ell(\beta) = n^{-1} \sum_{i=1}^n Y_i \log(P_i) + (1 - Y_i) \log(1 - P_i), \quad (16)$$

differing from each other only in the specification of $P_i \equiv \Pr(Y_i = 1 | X_i' \beta)$. The first parametric model we consider is a simple *homoskedastic logit* (L) model given by

$$P_i^L = L(X_i' \beta / \sigma) \equiv \frac{1}{1 + \exp(-X_i' \beta / \sigma)}. \quad (17)$$

In this model, the covariates exert a pure location effect on the latent utility differential. The second parametric model allows for (parametric) heterogeneity of the form $V(Y_i^* | Z_i) = \sigma^2 \exp(2Z_i' \gamma)$, which yields a *heteroskedastic logit* (HL) model given by

$$P_i^{HL} = L\left(\frac{X_i' \beta}{\sigma \exp(Z_i' \gamma)}\right). \quad (18)$$

In order to allow for possible violations of the logit link function, we also consider the *generalized heteroskedastic logit model* (GHL) proposed by Stukel (1988)

$$P_i^{GHL} = L\left[g_v\left(\frac{X_i' \beta}{\sigma \exp(Z_i' \gamma)}\right)\right], \quad (19)$$

where $g_v(\eta)$ is a strictly increasing function of η indexed by two shape parameters $v = (v_R, v_L)$ controlling respectively the left and right tails of the error distribution. For $\eta \geq 0$ ($P \geq 0.5$),

$$g_v(\eta) = \begin{cases} v_R^{-1}(\exp(v_R \eta) - 1), & v_R > 0 \\ \eta, & v_R = 0 \\ -v_R^{-1} \log(1 - v_R \eta), & v_R < 0, \end{cases}$$

while for $\eta < 0$ ($P < 0.5$),

$$g_v(\eta) = \begin{cases} -v_L^{-1}(\exp(v_L |\eta|) - 1), & v_L > 0 \\ \eta, & v_L = 0 \\ v_L^{-1} \log(1 - v_L |\eta|), & v_L < 0. \end{cases}$$

Stukel (1988) shows that this model can approximate both symmetric link functions like the probit link, as well as asymmetric ones like the log-log and the complementary log-log.

Finally, we also consider the semiparametric *single-index* (SI) model of Klein and Spady (1993). To define this estimator, let $k(\cdot)$ be a kernel function, h_n be a bandwidth parameter that converges to zero as n becomes large, and define the leave- i -out estimate of $\Pr(Y = 1|X'_i\beta)$ by

$$\tilde{P}_i = \sum_{j \neq i} Y_j k\left(\frac{X'_i\beta - X'_j\beta}{h_n}\right) / \sum_{j \neq i} k\left(\frac{X'_i\beta - X'_j\beta}{h_n}\right).$$

To avoid anomalous behaviour at the tails, let ε be a small positive number and define

$$P_i^{SI} = \max\{\varepsilon, \min\{1 - \varepsilon, \tilde{P}_i\}\}. \quad (20)$$

Klein and Spady (1993) have shown that, under some additional regularity conditions, if $nh_n^6 \rightarrow \infty$ and $nh_n^8 \rightarrow 0$ as $n \rightarrow \infty$, the estimator converges at the parametric root- n rate to an asymptotically normal random variable, and achieves the semiparametric efficiency bound.

The logit and single-index models were estimated by maximum likelihood in **Stata**. The bandwidth parameter in the single-index model was chosen by the cross-validation procedure proposed by Hardle *et al.* (1993). The quantile models were computed using the Goffe *et al.* (1994) simulated annealing algorithm in **Fortran** and **S**. The bandwidth for the quantile estimators was chosen using Horowitz's (1992) plug-in method. All software is available from the author upon request.

5.2. The Data

The data were extracted from the 1999 March Supplement to the US Current Population Survey. In selecting the sample and appropriate explanatory variables, we follow Olson (1998). To create the sample, all the primary, single-family households were extracted from the data set. Next, we dropped all families for which the wife was below 20, or above 65 years of age, as well as families

Table I. Descriptive statistics

Variable	Definition	Mean	Std dev.	Min	Max
LFP	Indicator of participation	0.728	0.445	0	1
EDUC	(Years of schooling)/10	1.368	0.285	0	2.2
EXP	Potential labour force experience = (AGE - EDUC - 6)/10	2.298	1.146	-0.4	5.9
KIDSLT6	No. of children <6 years old	0.316	0.634	0	6
KIDS618	No. of children between 6 and 18 years old	0.707	0.998	0	8
NC	Indicator for residency in the North or Central USA	0.435	0.496	0	1
HHI	Indicator for wife's coverage by husband's health insurance	0.474	0.499	0	1
BLACK	Indicator for black	0.059	0.235	0	1
HISPANIC	Indicator for Hispanic origin	0.140	0.347	0	1
LINC	Log(non-labour income) = log of total family income excluding wife's earnings (in \$10 000)	1.355	0.796	-2.408	4.021

$N = 20\,871$ observations.

Source: 1999 March Supplement to the US Current Population Survey.

for which the relevant income variables were assigned allocative values. Finally, all families for which the wife's labour force status changed since the previous year were also dropped. These operations produced a sample of 20 871 families. The variable we wish to model is the labour force participation (LFP) decision of the wife, defined as 1 if her usual hours worked were non-zero, and 0 otherwise. Table I gives definitions and reports descriptive statistics for the variables in the model.

5.3. Estimation Results

Comparison of Logit with the Median and Smoothed Median Estimates

Table II presents estimation results for the parametric models in three panels. The *location* panel reports estimates of β , the *scale* panel presents estimates of γ , while the *tails* panel reports estimates of ν . Likelihood ratio (LR) test statistics for heteroskedasticity and logistic tails are also presented.

To facilitate comparisons of the *location* models it is convenient to impose the $\|\beta\| = 1$ normalization. Table III presents the location models of the three parametric specifications alongside the semiparametric single-index, median and smoothed median models. Looking first at the three parametric models, we see that all coefficients have the expected sign and are highly significant, but there are considerable discrepancies in the magnitudes of the point estimates. Years of schooling (EDUC) has the expected positive effect on the probability of participation, experience (EXP) exerts a significant quadratic effect, and non-labour income (LNLINC) has the theoretically expected negative effect. The coefficients of two ethnicity variables BLACK and HISPANIC have opposite signs, with BLACK having a positive and HISPANIC a negative effect on the probability of participation. The number of children has the expected negative effect, with young children (KIDSLT6) having a more pronounced effect than older children (KIDS618). The indicator for residency in the North and Central USA (NC) has a positive effect on the probability of participation, while the indicator of wife's coverage by husband's health insurance (HHI) has the expected negative effect. The magnitudes of these point estimates vary considerably among the models, with the coefficient of EDUC in the HL model being 32% smaller than that in model L, and the coefficient of LNLINC in the GHL model being 33% larger than that in model L. Looking next at the three semiparametric single-index and median models in Table III, we see that all variables are again statistically significant and have the expected signs, but here too there are considerable discrepancies in the magnitude of the point estimates, both among the semiparametric models and between them and their parametric counterparts.

The sensitivity of the parametric estimates to the specification of the model suggests distributional heterogeneity in the data. In fact, looking at the *scale* functions of the two heteroskedastic models in Table II, we see that the variables EDUC, HISPANIC, LNLINC, LNLINC² and HHI enter the function in a statistically significant way. The LR statistic of about 257 on nine degrees of freedom strongly rejects the homoskedasticity assumption. In particular, EDUC is seen to have a doubly significant effect on WTP: more educated women not only have a higher WTP (location effect), but also have less variability in their choice than less educated women (scale effect). Non-labour (LNLINC) has a convex effect on scale, with women with very high and very low non-labour income having more variability in their choice. Women of Hispanic origin (HISPANIC) may have a lower WTP (negative location effect), but also have a significantly higher variability. Finally, husband's health insurance (HHI) is shown to have a positive effect on the willingness

Table II. Parametric and flexible parametric models of the labour force participation decision of married women (normalization: $\sigma = \pi/\sqrt{3}$, the standard deviation of the logistic distribution)

	Variable	Homoskedastic logit	Heteroskedastic logit	Generalized heteroskedastic logit
Location	CONSTANT	0.274 [†] (0.139)	0.254 [‡] (0.063)	0.193 [†] (0.086)
	EDUC	1.183 [‡] (0.074)	0.253 [‡] (0.090)	0.310 [†] (0.123)
	EXP	0.731 [‡] (0.068)	0.259 [‡] (0.073)	0.333 [‡] (0.108)
	EXP ²	-0.265 [‡] (0.013)	-0.094 [‡] (0.021)	-0.112 [‡] (0.030)
	BLACK	0.291 [‡] (0.078)	0.116 [†] (0.057)	0.121 (0.065)
	HISPANIC	-0.329 [‡] (0.051)	-0.095 [‡] (0.027)	-0.115 [‡] (0.035)
	KIDSLT6	-0.759 [‡] (0.030)	-0.251 [‡] (0.046)	-0.275 [‡] (0.063)
	KIDS618	-0.189 [‡] (0.019)	-0.067 [‡] (0.016)	-0.083 [‡] (0.022)
	LINC	-0.125 [‡] (0.024)	-0.048 [‡] (0.018)	-0.059 [‡] (0.022)
	HHI	-0.519 [‡] (0.036)	-0.142 [‡] (0.032)	-0.155 [‡] (0.042)
	NC	0.283 [‡] (0.035)	0.101 [‡] (0.029)	0.109 [‡] (0.036)
Scale	EDUC	—	-0.690 [‡] (0.098)	-0.964 [‡] (0.144)
	EXP	—	-0.049 (0.030)	0.022 (0.050)
	BLACK	—	0.074 (0.099)	0.026 (0.111)
	HISPANIC	—	0.197 [‡] (0.075)	0.255 [‡] (0.087)
	KIDSLT6	—	-0.034 (0.036)	0.078 (0.069)
	KIDS618	—	-0.001 (0.024)	0.002 (0.030)
	LINC	—	-0.373 [‡] (0.058)	-0.442 [‡] (0.071)
	LINC ²	—	0.148 [‡] (0.019)	0.185 [‡] (0.025)
	HHI	—	0.105 [†] (0.045)	0.206 [‡] (0.060)
	NC	—	0.007 (0.046)	-0.041 (0.054)
Tails	ν_R	—	—	-0.335 [‡] (0.102)
	ν_L	—	—	-0.112 (0.194)
	Log-likelihood	-10 666.6	-10 538.0	-10 530.3
	LR statistic		257.2 [‡]	15.4 [‡]
	χ^2 d.f.		9	2

Asymptotic standard errors in parentheses.

[†] indicates significance at the 5% level.[‡] indicates significance at the 1% level.

variance function, so that women that are *not* covered by their husband's insurance are not only more willing to work but are also more consistent in their choice. Comparing the GHL with the HL model, we see that the HL model is rejected by the LR test, and the statistically significant

Table III. Comparison of parametric estimates with median and smoothed median estimates of the labour force participation decision of married women (normalization: $\beta_{\text{LINC}} = -1$)

Variable	Homoskedastic logit	Heteroskedastic logit	Generalized heteroskedastic logit	Semiparametric single index	Median maximum score	Smoothed median maximum score
CONSTANT	2.184 (1.229)	5.255 (2.692)	3.282 (2.232)	—	-2.267 [‡] [-2.757, -1.495]	-4.022 [†] (1.693)
EDUC	9.444 [‡] (1.759)	5.240 [‡] (1.854)	5.278 [‡] (1.808)	8.645 [‡] (1.315)	12.517 [‡] [11.208, 15.768]	10.720 [‡] (2.001)
EXP	5.832 [‡] (1.171)	5.356 [‡] (1.705)	5.663 [‡] (1.757)	5.522 [‡] (0.917)	6.050 [‡] [5.455, 7.694]	5.833 [‡] (0.762)
EXP ²	-2.116 [‡] (0.410)	-1.945 [‡] (0.619)	-1.908 [‡] (0.574)	-1.895 [‡] (0.299)	-2.150 [‡] [-2.746, -1.943]	-1.878 [‡] (0.326)
BLACK	2.320 [‡] (0.795)	2.399 (1.241)	2.063 (1.080)	1.682 [‡] (0.542)	-0.167 [-0.828, 8.215]	-0.105 (0.086)
HISPANIC	-2.628 [‡] (0.639)	-1.973 [†] (0.887)	-1.953 [†] (0.808)	-2.515 [‡] (0.462)	-3.417 [†] [-3.626, 0.660]	-2.755 [‡] (0.489)
KIDSLT6	-6.062 [‡] (1.204)	-5.202 [‡] (1.764)	-4.677 [‡] (1.470)	-4.826 [‡] (0.783)	-5.567 [‡] [-7.766, -5.070]	-4.241 [‡] (1.139)
KIDS618	-1.512 [‡] (0.332)	-1.378 [‡] (0.473)	-1.415 [‡] (0.451)	-1.373 [‡] (0.236)	-1.783 [‡] [-2.408, -1.520]	-1.306 [‡] (0.333)
LINC	-1.000 (—)	-1.000 (—)	-1.000 (—)	-1.000 (—)	-1.000 [—]	-1.000 (—)
HHI	-4.145 [‡] (0.925)	-2.933 [‡] (1.048)	-2.631 [‡] (0.893)	-3.222 [‡] (0.609)	-4.867 [‡] [-5.894, -3.210]	-3.823 [‡] (0.814)
NC	2.260 [‡] (0.515)	2.085 [‡] (0.769)	1.856 [‡] (0.650)	1.765 [‡] (0.348)	2.033 [‡] [1.485, 3.373]	1.437 [‡] (0.469)

Asymptotic standard errors in parentheses and 95% bootstrap confidence intervals in square brackets.

[†] indicates significance at the 5% level.

[‡] indicates significance at the 1% level.

negative value of the ν_R parameter indicates a skewed-right error distribution (see Stukel, 1988 for details on the interpretation of these tail parameters).

Smoothed Binary Regression Quantiles

While the results of the parametric models yield a very informative and plausible structure for the labour force participation decision of married women, their validity depends crucially on potentially restrictive specification assumptions. At the other extreme, the median models may be consistent under quite general conditions but do not provide any information about potential heterogeneity in the sample. To obtain a more complete picture we estimate a family of binary quantile regression models over the grid $\theta = \{0.10, 0.15, \dots, 0.65\}$. To justify this choice, recall that the unconditional probability of participation in this US sample is around 73%, which leads us to expect that a model around the 0.30 conditional quantile would be the one most efficiently estimable. Given that high conditional quantiles of WTP correspond to low conditional probabilities of participation, the high unconditional probability of participation in the USA means that there are very few American women that belong to very high conditional quantiles of WTP.

The smoothed binary quantile estimates along with their 95% asymptotic confidence intervals are graphed in Figure 1. We see that EDUC receives an increasing weight as we move from low towards higher quantiles, indicating that education exerts a stronger influence in the upper tail

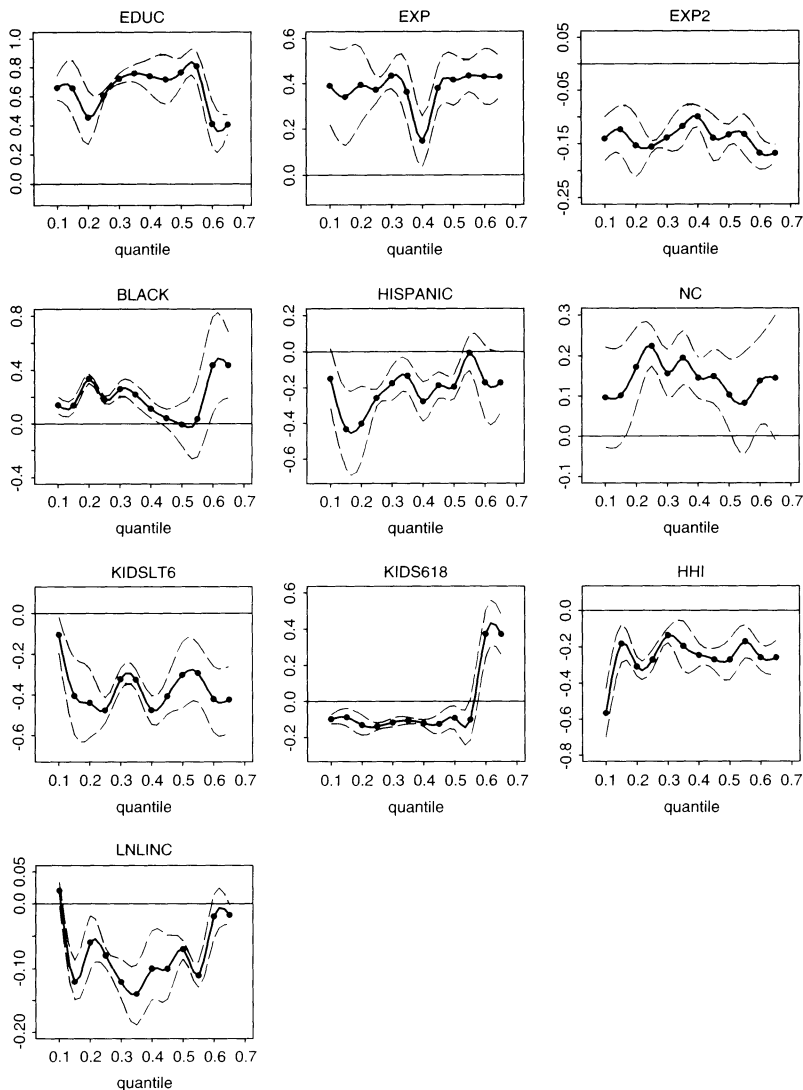


Figure 1. Smoothed binary regression quantile estimates and 95% confidence intervals of the labour force participation decision of married women (normalization: $\|\beta_{n\tau}^{-c}\| = 1$)

of the WTP distribution. At the 0.60 quantile, however, there is a precipitous drop in the EDUC effect, which may be attributable to the corresponding effect of the number of older children which becomes positive at those quantiles. The number of older children, KIDS618, has a rather stable, statistically significant, negative effect on the decision to participate up to the 0.60 quantile, at which point the coefficient of this variable becomes, and remains from there on, significantly positive. On the contrary, the number of young children, KIDSLT6, is significantly negative at all quantiles. This interesting reversal of signs of the KIDS618 variable suggests that women that are

high in the willingness distribution may decide not to participate when they have young children, but will regard older children as a positive incentive to work. EXP has across all quantiles a stable quadratic effect on participation. Furthermore, HISPANIC receives a negative weight, while BLACK is significantly positive at quantiles below the median. NC has a positive effect on participation, while HHI has a significantly negative effect across all quantiles. Finally, LNLINC has a quadratic in τ effect, being negative and significant at the middle quantiles, but receiving smaller weight and becoming insignificant at very low and very high quantiles. This leads us to believe that non-labour income is not a significant consideration in the decision of women who have strong preferences either for or against working.

Figure 2 presents the results of the non-parametric specification test outlined in Section 4.3. For all quantile models considered here, the non-parametric estimate evaluated at zero is not significantly different from $(1 - \tau)$ at the 5% level, leading us to conclude that these models constitute a reasonable approximation of the true quantile function of the latent utility differential.

5.4. Probabilities

The three parametric models, the single-index model and the family of smoothed quantile models may be compared in terms of their probability predictions. In order to evaluate the performance of the models globally and avoid a large number of examples that require specifying and reporting individual characteristics, we adopt the following strategy. For each of the 12 quantile models estimated here, we compute the index $X'_i b_{n\tau} / \|b_{n\tau}\|$, $i = 1, \dots, n$. Given a small positive number δ , we then create as many strata (subsamples) as there are quantile estimates, with an individual i belonging to stratum τ if $-\delta \leq X'_i b_{n\tau} / \|b_{n\tau}\| \leq \delta$, so that i 's participation probability according to the quantile models is approximately $(1 - \tau)$. Here δ acts like a bandwidth parameter, with larger values relaxing, and smaller values making more strict, the requirement that the computed index is close to zero. Given these strata of women with, according to the quantile models, a fixed participation probability, the fitted probabilities for the three parametric and the single-index models are computed and subsample statistics (across all women in these strata) are obtained.

Table IV presents sample statistics from these 12 strata. The SBRQ column of the table reports the estimation quantile τ , the implied participation probability $\hat{P}$ and the size of the subsample of women that belong to each stratum $n_\tau(\delta)$, for each of the two values of $\delta = \{0.05, 0.01\}$ considered here. The rest of the columns in the table report the corresponding means and standard deviations of predicted probabilities for each of the parametric and single-index models in each of the 12 subsamples implied by the quantile models. We are interested in comparing the mean predicted probabilities $\mu(\hat{P})$ among the models, the standard error of which may be approximated by $\sigma(\hat{P}) / \sqrt{n_\tau(\delta)}$.³

For quantiles between 0.20 and 0.45 corresponding to participation probabilities between 55% and 85%, the three logit models agree closely with each other and with the semiparametric single-index and smoothed quantile regression models. Given the 73% unconditional probability of participation in this sample, it is reasonable to say that this 55% to 85% probability area corresponds to a group of women that behave more or less homogeneously, and that their behaviour is adequately characterized by all the models considered here. Outside this range, however, the

³ This standard error does not take into account the variability of P due to estimation of the coefficient vector, but since these are means over quite large subsamples, this latter variation should be of second-order to the variation over the subsample.

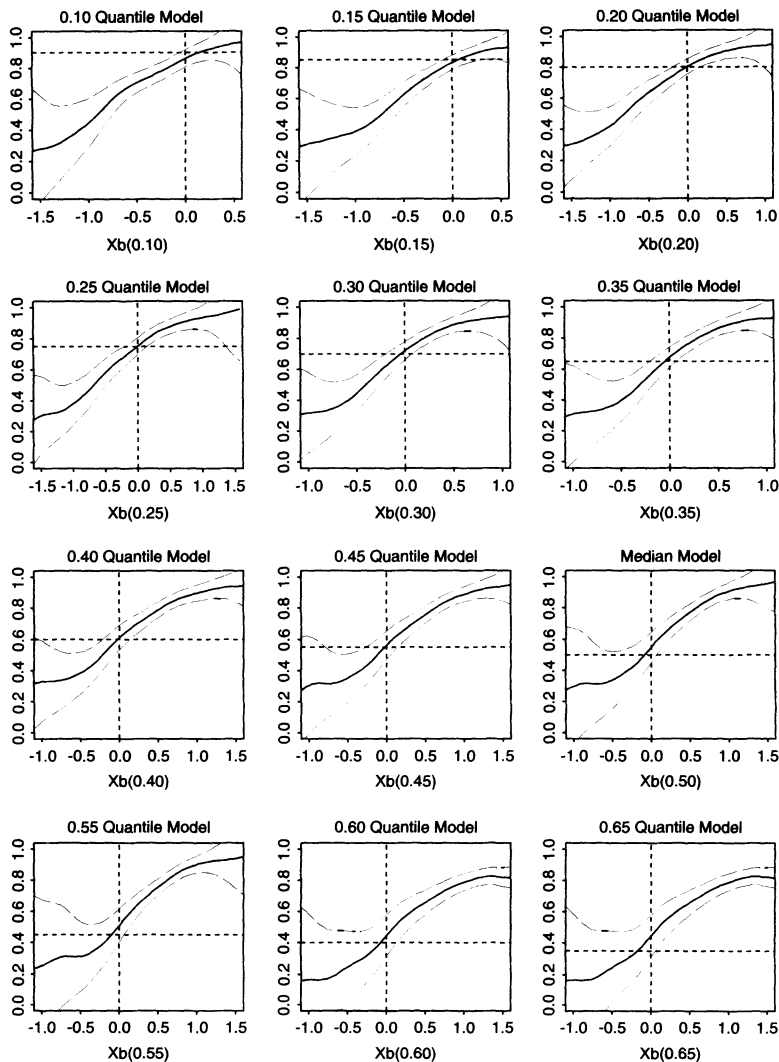


Figure 2. Specification tests of binary regression quantile models: nonparametric estimates and 95% uniform confidence bands

models disagree considerably. In particular, for women that according to the quantile models have predicted participation probabilities above 85%, participation probabilities are *under*-predicted by the three parametric and the single-index models, with the severity of under-prediction being the largest when the simple logit model is used and becoming smaller for the two heteroskedastic models and the single-index model. The disagreements become more pronounced when we consider women that according to the quantile models have participation probabilities below 55%. For this group of women, the three parametric models and the single-index model *over*-predict the participation probability, with the degree of over-prediction being the largest for the simple logit

Table IV. Predicted choice probabilities

	Smoothed binary regression quantiles			Homoskedastic logit		Heteroskedastic logit		Generalized heteroskedastic logit		Semiparametric single index	
	τ	$\hat{p}^{SBRQ}$	$n_\tau(\delta)$	$\mu(\hat{p}^L)$	$\sigma(\hat{p}^L)$	$\mu(\hat{p}^{HL})$	$\sigma(\hat{p}^{HL})$	$\mu(\hat{p}^{GHL})$	$\sigma(\hat{p}^{GHL})$	$\mu(\hat{p}^{SI})$	$\sigma(\hat{p}^{SI})$
$\delta = 0.05$	0.10	0.90	1832	0.868	0.050	0.878	0.063	0.878	0.060	0.878	0.058
	0.15	0.85	2397	0.833	0.020	0.840	0.039	0.841	0.038	0.843	0.031
	0.20	0.80	1730	0.803	0.023	0.804	0.044	0.806	0.045	0.806	0.034
	0.25	0.75	1619	0.759	0.019	0.754	0.039	0.758	0.042	0.751	0.029
	0.30	0.70	2109	0.717	0.035	0.707	0.038	0.712	0.039	0.700	0.039
	0.35	0.65	1723	0.673	0.048	0.658	0.051	0.662	0.050	0.659	0.050
	0.40	0.60	1012	0.606	0.034	0.589	0.043	0.589	0.044	0.602	0.045
	0.45	0.55	783	0.559	0.024	0.544	0.038	0.539	0.038	0.545	0.049
	0.50	0.50	742	0.542	0.042	0.525	0.057	0.517	0.059	0.520	0.058
	0.55	0.45	672	0.493	0.051	0.481	0.062	0.470	0.063	0.458	0.062
	0.60	0.40	346	0.436	0.072	0.417	0.040	0.406	0.041	0.403	0.065
	0.65	0.35	333	0.435	0.073	0.417	0.042	0.407	0.043	0.403	0.065
$\delta = 0.01$	0.10	0.90	407	0.870	0.046	0.883	0.057	0.882	0.054	0.881	0.055
	0.15	0.85	453	0.833	0.018	0.842	0.031	0.843	0.031	0.843	0.030
	0.20	0.80	361	0.804	0.022	0.807	0.044	0.809	0.045	0.810	0.032
	0.25	0.75	333	0.759	0.017	0.751	0.039	0.754	0.042	0.752	0.022
	0.30	0.70	419	0.714	0.032	0.705	0.037	0.711	0.038	0.695	0.039
	0.35	0.65	330	0.672	0.044	0.657	0.047	0.661	0.045	0.655	0.048
	0.40	0.60	205	0.606	0.032	0.593	0.041	0.592	0.042	0.606	0.040
	0.45	0.55	141	0.556	0.021	0.538	0.038	0.534	0.037	0.538	0.040
	0.50	0.50	136	0.544	0.043	0.529	0.056	0.522	0.057	0.522	0.051
	0.55	0.45	127	0.491	0.047	0.483	0.054	0.473	0.054	0.450	0.059
	0.60	0.40	49	0.428	0.087	0.419	0.058	0.412	0.052	0.407	0.062
	0.65	0.35	47	0.430	0.071	0.416	0.058	0.408	0.055	0.406	0.054

model and the smallest for the generalized heteroskedastic logit model. This pattern of under- and over-prediction may be interpreted as evidence of the restrictiveness of the parametric and single-index models that tend to extrapolate the behaviour of women around the mean to women that belong to the tails of the conditional WTP distribution.

APPENDIX A: OUTLINE PROOF OF THEOREM 2

This Appendix contains an outline of the proof of Theorem 2. Complete proofs of this and the other theorems stated in the main text are available from the author upon request. The proof of Theorem 2 is a generalization of the proof of theorem 2 of Horowitz (1992), first to general quantiles and then to a family of quantile estimates.

(a) From Theorem 1, for each τ_j , $T_{n\tau_j}(b_{n\tau_j}; h_n) = 0$ with probability 1, so a Taylor series expansion yields

$$T_{n\tau_j}(\beta_{\tau_j}; h_n) + H_{n\tau_j}(b^*; h_n)(\widetilde{b}_{n\tau_j} - \widetilde{\beta}_{\tau_j}) = 0, \tag{A1}$$

with $T_{n\tau_j}$ and $H_{n\tau_j}$ as defined in (9) and (10) and b^* between β_{τ_j} and $b_{n\tau_j}$. The proof of part (a) will now be given in three steps.

Step 1. We first show that $\text{plim}_{n \rightarrow \infty} H_{n\tau_j} = H_{\tau_j}$. From the arguments in lemma 9 of Horowitz (1992), $H_{n\tau_j}$ converges in probability to its expectation given by

$$EH_{n\tau_j} = h_n^{-2} \int [\tau - F_{\tau_j}(-Z_{\tau_j}|\tilde{X})] \tilde{X} \tilde{X}' K^{(2)}(Z_{\tau_j}/h_n) p(Z_{\tau_j}|\tilde{X}) dZ_{\tau_j} dP(\tilde{X}). \quad (\text{A2})$$

Expanding $F_{\tau_j}(-Z_{\tau_j}|\tilde{X})$ and $p_{\tau_j}(Z_{\tau_j}|\tilde{X})$ about $Z_{\tau_j} = 0$, and using $F_{\tau_j}(0|\tilde{X}) = \tau_j$ we obtain

$$[\tau_j - F_{\tau_j}(-Z_{\tau_j}|\tilde{X})] \cdot p_{\tau_j}(Z_{\tau_j}|\tilde{X}) = -F_{\tau_j}^{(1)}(0|\tilde{X}) p_{\tau_j}(0|\tilde{X}) Z_{\tau_j} + o_p(1). \quad (\text{A3})$$

Combining (A2) and (A3) yields

$$EH_{n\tau_j} = -h_n^{-2} \int \tilde{X} \tilde{X}' F_{\tau_j}^{(1)}(0|\tilde{X}) p(0|\tilde{X}) Z_{\tau_j} K^{(2)}(Z_{\tau_j}/h_n) dZ_{\tau_j} dP(\tilde{X}) + o_p(1)$$

which converges to H_{τ_j} by Lebesgue's dominated convergence theorem.

Step 2. Next we show that as $n \rightarrow \infty$, if $nh_n^{2s+1} \rightarrow \lambda$, a finite positive number

$$(nh_n)^{1/2} T_{n\tau_j}(\beta_{\tau_j}; h_n) \xrightarrow{d} \mathcal{N}_{k-1}(\lambda^{1/2} A_{\tau_j}, D_{\tau_j}). \quad (\text{A4})$$

Let $T_{n\tau_j} \equiv T'_{n\tau_j}(\beta_{\tau_j}; h_n)$ and write

$$(nh_n)^{1/2} T_{n\tau_j} = (nh_n)^{1/2} [T_{n\tau_j} - ET_{n\tau_j}] + (nh_n)^{1/2} ET_{n\tau_j}. \quad (\text{A5})$$

Write the second term in (A5) as $(nh_n)^{1/2} ET_{n\tau_j} = (nh_n^{2s+1})^{1/2} E h_n^{-s} T_{n\tau_j}$ and note that under Assumptions 5–7,

$$E h_n^{-s} T_{n\tau_j} = h_n^{-s} \int [\tau - F_{\tau}(-Z_{\tau_j}|\tilde{X})] K^{(1)}(Z_{\tau_j}/h_n) (\tilde{X}/h_n) p_{\tau_j}(Z_{\tau_j}|\tilde{X}) dZ_{\tau_j} dP(\tilde{X}). \quad (\text{A6})$$

Under Assumption 3, $F_{\tau_j}(0|\tilde{X}) = \tau_j$ a.s. $P(\tilde{X})$. Let $W_{\tau_j} = Z_{\tau_j}/h_n$ and expand $[\tau_j - F_{\tau_j}(-h_n W_{\tau_j}|\tilde{X})]$ and $p_{\tau_j}(h_n W_{\tau_j}|\tilde{X})$ about $h_n W_{\tau_j} = 0$ to get

$$\begin{aligned} & [\tau_j - F_{\tau_j}(-h_n W_{\tau_j}|\tilde{X})] p_{\tau_j}(h_n W_{\tau_j}|\tilde{X}) \\ &= - \left[\frac{1}{s!} F_{\tau_j}^{(s)}(-\xi_s|\tilde{X}) p_{\tau_j}(h_n W_{\tau_j}|\tilde{X}) + \sum_{i=1}^{s-1} \frac{1}{i!(s-i)!} F_{\tau_j}^{(i)}(0|\tilde{X}) p_{\tau_j}^{(s-i)}(\xi_i|\tilde{X}) \right] h_n^s W_{\tau_j}^s \\ &\quad - \sum_{i=1}^{s-1} \sum_{k=0}^{s-i-1} \frac{1}{i!k!} F_{\tau_j}^{(i)}(0|\tilde{X}) p_{\tau_j}^{(k)}(0|\tilde{X}) h_n^{i+k} W_{\tau_j}^{i+k} \end{aligned} \quad (\text{A7})$$

for $\xi_1, \dots, \xi_s$ scalars with values between 0 and $h_n W_{\tau_j}$. Analogously to lemma 5 of Horowitz (1992), by Assumptions 5 and 6 there exists an $\eta > 0$ such that the derivatives exist and are uniformly bounded for almost every $\tilde{X}$ if $|h_n W_{\tau_j}| \leq \eta$. Substitution of (A7) into (A6) and Lebesgue's dominated convergence yields $E h_n^{-s} T_{n\tau_j} \rightarrow A_{\tau_j}$, and thus $E(nh_n)^{1/2} T_{n\tau_j} \rightarrow \lambda^{1/2} A_{\tau_j}$.

Turning to the first term in (A5), arguments similar to those in lemma 6 of Horowitz (1992) can be used to show that the i.i.d. zero-mean random element $(nh_n)^{1/2}[T_{n\tau_j} - ET_{n\tau_j}]$ is asymptotically normal. To derive the asymptotic covariance matrix write

$$T_{n\tau_j} = n^{-1} \sum_{i=1}^n t_{ni\tau_j}(b_{n\tau}; h_n)$$

for $t_{ni\tau_j}(\beta_{n\tau_j}; h_n) \equiv h_n^{-1}[Y_i - (1 - \tau)]\tilde{X}_i K^{(1)}(X'_i \beta_{n\tau_j}/h_n)$. Now

$$\begin{aligned} \text{Var}[(nh_n)^{1/2}T_{n\tau_j}(\beta_{\tau_j}; h_n)] &= h_n E(t_{n\tau_j} t'_{n\tau_j}) - h_n E[(nh_n)^{1/2}T_{n\tau_j}] E[(nh_n)^{1/2}T_{n\tau_j}]' \\ &= h_n E(t_{n\tau_j} t'_{n\tau_j}) + o_p(1) \\ &= \tau_j(1 - \tau_j) h_n^{-1} \int [K^{(1)}(Z_{\tau_j}/h_n)]^2 \tilde{X} \tilde{X}' p_{\tau_j}(Z_{\tau_j}|\tilde{X}) dZ_{\tau_j} dP(\tilde{X}) + o_p(1) \\ &= \tau_j(1 - \tau_j) \int [K^{(1)}(W_{\tau_j})]^2 \tilde{X} \tilde{X}' p_{\tau_j}(h_n W_{\tau_j}|\tilde{X}) dW_{\tau_j} dP(\tilde{X}) + o_p(1) \end{aligned}$$

which converges to D_{τ_j} by Lebesgue's dominated convergence. Using Assumption 7, we now obtain the following asymptotic linearity relation:

$$(nh_n)^{1/2}(\tilde{b}_{n\tau_j} - \tilde{\beta}_{\tau_j}) = -H_{\tau}^{-1}(nh_n)^{1/2}T_{n\tau_j}(\beta_{\tau_j}; h_n) + o_p(1). \quad (\text{A8})$$

The result in (A4) follows by an application of the Lindeberg–Feller central limit theorem.

Step 3. The final step is to derive the asymptotic covariance matrix between two regression quantiles β_{τ_j} and β_{τ_ℓ} , i.e., show that as $n \rightarrow \infty$,

$$\text{Cov}[(nh_n)^{1/2}T_{n\tau_j}(\beta_{\tau_j}; h_n), (nh_n)^{1/2}T_{n\tau_\ell}(\beta_{\tau_\ell}; h_n)] \xrightarrow{p} \tilde{D}_{\tau_j\tau_\ell} \quad (\text{A9})$$

where $\tilde{D}_{\tau_j\tau_\ell}$ is the $(k-1) \times (k-1)$ matrix defined in (8). We have

$$\begin{aligned} \text{Cov}[(nh_n)^{1/2}T_{n\tau_j}, (nh_n)^{1/2}T_{n\tau_\ell}] &= h_n E(t_{n\tau_j} t'_{n\tau_\ell}) + o_p(1) \\ &= (\min\{\tau_j, \tau_\ell\} - \tau_j\tau_\ell) \cdot h_n^{-1} \int K^{(1)}(Z_{\tau_j}/h_n) K^{(1)}(Z_{\tau_\ell}/h_n) \\ &\quad \tilde{X} \tilde{X}' p_{\tau_j}(z_{\tau_j}|\tilde{X}) p_{\tau_\ell}(Z_{\tau_\ell}|\tilde{X}) dZ_{\tau_j} dZ_{\tau_\ell} dP(\tilde{X}) + o_p(1) \end{aligned}$$

and Lebesgue's dominated convergence theorem yields (A9).

(b) From part (a), $\text{AMSE} = \text{trace}[\lambda^{-1/(2s+1)} \Sigma_\theta + \lambda^{2s/(2s+1)} \Lambda_\theta \Lambda'_\theta]$. Differentiating the right-hand side with respect to λ yields that AMSE is minimized by setting $\lambda = \lambda_\theta^*$, in which case it follows from part (a) that $n^{s/(2s+1)}(\tilde{c}_{n\theta} - \tilde{\gamma}_\theta)$ has the specified asymptotic distribution. $\square$

ACKNOWLEDGEMENTS

I am grateful to Roger Koenker for numerous insightful comments and suggestions. Many thanks are also due to George Deltas, the editor, and three anonymous referees for their comments that have greatly improved the substance and presentation of the paper.

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